

Cashton Winpigler

Professor Miller

CSIT 203

28 January 2026

Miniaturization of computer technology

Is miniaturization a constant force throughout the history of computer technology?

Miniaturization is a constant force in the history of computer technology and will continue.

Miniaturization of computers has happened in the past and will happen in the future.

One reason that miniaturization is a constant force in computer technology is because simply the size of computers has decreased over time. The ENIAC in the 1940's took up 1800 square feet, while a computer of today's age will fit onto a desk. Another reason computers have gotten smaller is that it lets computers be more efficient and have more things inside them to boost performance. Overall, in the last few decades computer technology has physically gotten smaller.

Computers to program and run have also gotten faster, meaning the time it takes for a computer to operate has also been miniaturized. The oldest of computers require a week worth of set up or programming, while modern computers require much less time to program, making programs that use to take a week to set up in only a few hours. Computers have also been optimized by letting them run faster and have more things running at once, while older computers take days to run programs, a modern computer can take seconds or hours for

complex tasks. In summary, computers have gotten faster, reducing the amount of time it takes to do something with a computer.

The future of computer technology also maintains high hopes for the miniaturization of computer technology with Quantum Computing. Modern quantum computers are massive, taking up rooms for cooling purposes, but in the future as the cooling is handled and optimized, the size of them will be greatly reduced. Another part of quantum computers is that they are dramatically faster than the average computer, meaning the time it takes to run certain programs will also be shrunk. Overall, the future of computer technology is leading to even smaller and faster computers.

In summary, computer technology has always been affected by miniaturization. The size of computers has decreased by a great deal. Also, the time it takes to operate a computer and program one has been greatly reduced. Finally, the future of computer technology looks even smaller and faster with Quantum Computing being in the possibilities. Overall, the past, present, and future of computing will always be influenced by miniaturization.

I did use the internet for this to look up facts and numbers, but no AI.

<https://en.wikipedia.org/wiki/ENIAC>